

Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$

$AB = AC$ (By Ex-2.1, Prob 10)

$\det A = 2(6) - (-3)(-4) = 0$. By Theorem 4, A is not invertible. $\square$
But $B \neq C$.

14) Suppose $(B-C)D = O$, where B and C are $m \times n$ matrices and D is invertible. Show that $B = C$.

Solution: By Defn. of invertible matrix, there exists a unique matrix D^{-1} s.t.

$$D^{-1}D = DD^{-1} = I.$$

$$(B-C)D = O \Rightarrow ((B-C)D) \cdot D^{-1} = O \cdot D^{-1} \Rightarrow (B-C)(DD^{-1}) = O \cdot D^{-1} \text{ (Theorem 2(d))}$$

$$\Rightarrow (B-C)I = O \cdot D^{-1} \text{ (By defn. of inverse)} \Rightarrow B-C = O \cdot D^{-1} \text{ (Theorem 2(e))}$$

$$\Rightarrow (B-C)I = O \cdot D^{-1} \text{ (By defn. of zero matrix)} \Rightarrow (B-C) + C = O + C$$

$$\Rightarrow B + (-C + C) = O + C \text{ (Theorem 1(b))}$$

Let $C = [c_1 \ c_2 \ \dots \ c_n]$, where $1 \leq i \leq n$, $c_i \in \mathbb{R}^m$.

$\therefore -C = [-c_1 \ -c_2 \ \dots \ -c_n]$ (By Defn. of scalar multiplication)

$$-C + C = [-c_1 \ -c_2 \ \dots \ -c_n] + [c_1 \ c_2 \ \dots \ c_n]$$

$$= [-c_1 + c_1 \ -c_2 + c_2 \ \dots \ -c_n + c_n] \text{ (Defn. of matrix addition)}$$

$$= [0 \ 0 \ \dots \ 0] \text{ (Algebraic property in Sec-1.3)}$$

$$= O$$

$$\Rightarrow B + O = O + C \Rightarrow \boxed{B = C} \text{ (Theorem 1(c)) } \square$$

15) Suppose A, B and C are invertible $n \times n$ matrices. Show that ABC is also invertible by producing a matrix D such that $(ABC)D = I$ and $D(ABC) = I$

Solution:

By the defn. of invertible matrices, there exists unique matrices A^{-1}, B^{-1} and C^{-1} such that $AA^{-1} = A^{-1}A = I$, $BB^{-1} = B^{-1}B = I$ and $CC^{-1} = C^{-1}C = I$.

Claim:

The matrix $D = C^{-1}B^{-1}A^{-1}$ is a solution of both the equations $(ABC)D = I$ and $D(ABC) = I$.